GIS-ML-Commodities-Network — Blueprint

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Build `GIS-ML-Commodities-Network` with one spine: free satellite observations of supply-chain nodes produce supply surprises before official data, network theory maps how those surprises propagate through concentrated supply chains, and the price response is measured conditional on the macro and geopolitical regime.

Research question and fit

The question: can free satellite observations of activity at supply-chain nodes (fields, flare sites, mines, smelters, ports) detect supply shocks before official statistics do, and does the price impact of those shocks depend on the macro and geopolitical regime?

Why this framing beats "model everything against everything":

- It turns your list into one causal chain: physical activity → supply surprise → price response. Rates, inflation, labor, geopolitics and weather enter as regimes or shock sources, not as a pile of regressors.
- It is testable. 2025 handed you three copper natural experiments: Kamoa-Kakula seismic flooding (May), El Teniente tunnel collapse (July), and the Grasberg mud rush (Sept 8).
- Freeport declared force majeure at Grasberg about two weeks after the incident, and copper rose 3–4% that day. That gap is exactly where an independent physical signal has value.
- It ends in an investable object: a supply-shock state that conditions risk and positioning, which is what desk and research-associate interviews probe.

Where it sits:

- **Thesis:** Layer 4 is a ready thesis chapter: a satellite-derived supply index as an observable driver inside a stochastic commodity price model, estimated by MCMC. Brent, corn and copper map onto the three modules; silicon and rare earths are extensions.

Architecture



atlas architecture · 5 layers, 5 input and output feeds

Read top to bottom: each layer hands the next one object, and the right column shows what enters or leaves at that layer. The highlighted layer holds the core result, the regime-dependent price response.

Commodity modules

One pipeline, three node sets. Start with corn: it has the best free ground truth of any commodity on Earth.

Module	Nodes (where you look)	Satellite signals	Ground truth to nowcast	Market series
Corn (build first)	Corn pixels in Iowa, Illinois, Nebraska, masked by USDA Cropland Data Layer (10 m since 2024)	NDVI/EVI curves from NASA HLS; weather; drought	USDA NASS county and state yields; WASDE yield estimate as the consensus to beat	Corn futures (daily via yfinance); Alpha Vantage `CORN` (monthly, long history)
Permian oil and gas	Basin polygon plus flare clusters	VIIRS Nightfire flare count and radiant heat (nightly CSV); Sentinel-5P NO2; night lights	EIA monthly crude and gas production by state	WTI, Waha vs Henry Hub basis, E&P equities
Copper (then rare earths, silicon)	~20 mine footprints from the global mining polygons; ~30 largest smelters; export ports	Sentinel-2 SWIR (B11/B12) heat anomalies at smelters; Sentinel-5P SO2; Sentinel-1/2 change on pits, tailings, stockpiles; foundation-model embeddings	Company quarterly production; ICSG monthly releases; dated disruption events	Copper futures; Alpha Vantage `COPPER` (monthly); FCX, SCCO, COPX

- **Why corn first:** labels are free and dense, the literature gives you benchmarks to beat, and every stage of the pipeline gets exercised once before the harder modules.
- **The copper test:** did any free signal move between the Grasberg incident (Sept 8, 2025) and the force-majeure announcement? Grasberg Block Cave is underground, so you watch surface proxies: concentrator heat, tailings flow, night lights, port loading. If the signal stayed flat, that limit is itself a publishable finding.
- **Commercial benchmark:** Earth-i's SAVANT index tracks copper and nickel smelter activity from satellites for paying clients. You are building a small, open, transparent analogue.

Macro and geopolitical regimes

The macro variables enter as the regime a supply shock lands in, not as extra regressors. That keeps the model small enough to estimate on monthly data.

Driver	Series and source	Mechanism into commodities
Interest rates	10-yr real yield, fed funds, 2s10s (FRED via USAFacts; Alpha Vantage `TREASURY_YIELD`, `FEDERAL_FUNDS_RATE`)	Cost of carrying inventory; high real rates reward destocking
US dollar	Broad dollar index (FRED)	Dollar-priced commodities get dearer for non-US buyers
Inflation	CPI, 10-yr breakevens (FRED; Alpha Vantage `CPI`, `INFLATION`)	Hedge demand; supply shocks pass through to food and energy CPI
Labor	BLS mining and farm employment and wages; strike events from news data	Strikes are supply shocks (Chilean copper); wages shift the cost curve
Geopolitics	Caldara–Iacoviello GPR (global, 44 country indices, daily recent index); Alpha Vantage `NEWS_SENTIMENT`; USITC DataWeb tariffs and imports	Export controls, resource nationalism, chokepoints; tariffs reroute network edges
Environment	ERA5 or gridMET weather, US Drought Monitor, ENSO	Crop yields; water limits at Atacama copper mines; Gulf hurricanes

How they enter the model:

1. Fit a 2–3 state hidden Markov model on monthly real yield, dollar, GPR and CPI surprise. Reuse the classifier from your regime/risk repo.
2. Estimate state-dependent local projections: commodity returns at horizons 1–12 months on the supply surprise, interacted with the regime. `lpirfs::lp_nl` in R does this directly.
3. The deliverable sentence reads: "a one-standard-deviation copper supply shock moves price by X% in regime A versus Y% in regime B," with confidence bands.

Methods

Every layer has a stated estimand, a method and a validation test. Machine learning carries layers 1–2, network theory carries layer 3, and econometrics carries layers 4–5.

1. Satellite feature extraction

Signal	Construction	Quality control
Crop vigor	NDVI = (B08 − B04)/(B08 + B04) and EVI on Sentinel-2 L2A; zonal median per AOI per scene; weekly median composite	Scene Classification Layer masks cloud, shadow, cirrus and snow; clear-pixel fraction kept as a feature
Smelter heat	Normalized hotspot indices on SWIR, (B12 − B11)/(B12 + B11), plus hot-pixel count above a reflectance threshold inside a facility buffer	Daytime only; saturation and cloud flags; per-site baseline
Smelter and basin emissions	Sentinel-5P SO2 and tropospheric NO2 column; buffer mean minus a background ring (enhancement)	Pixel qa_value filter; wind-insensitive monthly means
Flaring	VIIRS Nightfire detections inside the basin polygon; daily count and summed radiant heat	Temperature threshold separates flares from biomass fires (configurable)
Land change	Foundation-model embeddings (Prithvi, TerraMind, AlphaEarth) → PCA; year-over-year cosine distance as a change score	Same season and sensor across years

Structural breaks in each activity index are dated with PELT change-point detection (penalized likelihood), which turns the 2025 copper disruptions into testable event dates.

2. Nowcasting and forecasting

The common activity factor comes from a dynamic factor model in state-space form, estimated by EM and the Kalman filter. It handles mixed frequencies and missing satellite looks natively.

$$x_{i,t} = \lambda_i f_t + e_{i,t}, \quad f_t = \phi f_{t-1} + u_t$$

Model	Role	Detail
Seasonal naive and AR(1)	Benchmarks every model must beat	Same point-in-time features
Dynamic factor model	Common activity factor from many noisy indices	Kalman smoother; loadings show which nodes drive the factor
Gradient-boosted trees	Nonlinear nowcast with weather interactions	Quantile loss at τ = 0.1, 0.5, 0.9; SHAP for attribution
Hierarchical linear model	Partial pooling across counties or sites	Random intercepts by region; shrinks noisy small sites toward the group

Validation is rolling-origin with an embargo between train and test windows, so no future data leaks into features. Reported metrics are RMSE, MAE, pinball loss and interval coverage, with Diebold–Mariano tests against the benchmark. Split-conformal intervals give finite-sample coverage without distributional assumptions.

The model output that feeds the rest of the system is a standardized supply surprise:

$$s_t = \frac{\hat{y}_t - y^{\text{consensus}}_t}{\hat{\sigma}_t}$$

3. Network theory

Two graphs carry the supply chain. The trade graph is bipartite: supplier countries on one side, commodities on the other, edge weight w_ck = import value of commodity k from country c (USITC). The physical graph is directed: mine → smelter → port → importer, with capacity weights.

Metric	Definition	What it tells a portfolio manager
Supplier share and HHI	$HHI_k = \Sigma_c (w_{ck} / \Sigma_c w_{ck})^2$	How concentrated supply of commodity k is

Metric	Definition	What it tells a portfolio manager
Effective number of suppliers	$N_{eff} = 1 / HHI_k$	Concentration in plain units: "supply behaves like 2.3 equal suppliers"
Strength	$s_c = \Sigma_k w_{ck}$	Total dependence on one supplier country
Eigenvector centrality and PageRank	Leading eigenvector of the country projection (countries linked by shared commodities)	Systemic suppliers whose disruption hits many commodities at once
Betweenness centrality	Share of shortest mine-to-importer paths through a node	Chokepoint smelters and ports
Nestedness (NODF)	Overlap of supplier sets, ordered by degree	Hidden common exposure: niche commodities sourced from the same hubs as bulk ones
k-core	Maximal subgraph where every node has degree $\geq k$	The irreducible core of the supply network
Robustness curve	Share of import value lost as suppliers are removed, targeted by strength vs random	How fast supply collapses under export bans; area under the curve summarizes it
Geopolitically weighted exposure	$GR_k = \Sigma_c share_{ck} \times GPR_c$	Concentration weighted by each supplier's geopolitical risk

Shock propagation runs as Monte Carlo: each supplier fails with probability p_c (calibrated to GPR or a stated prior), losses flow through the graph, and the simulation returns a supply-loss distribution per commodity, a supply-side Value-at-Risk.

4. Price linkage and regimes

A Gaussian hidden Markov model on standardized real yield, policy-rate change, inflation and unemployment change labels macro regimes. The number of states is chosen by BIC, and states are decoded by Viterbi.

Price response uses state-dependent local projections (Jordà 2005; Ramey and Zubairy 2018), with Newey–West standard errors:

```
p_{t+h} - p_{t-1} = \sum_{S \in \{A,B\}} \mathbb{1}[S_{t-
1}=S]\,\big(\alpha_{S,h} + \beta_{S,h}\, s_t + \gamma_{S,h}' x_t\big) +
\varepsilon_{t+h}
```

The impulse responses β_{S,h} answer the research question directly. Event studies on dated disruptions (cumulative abnormal returns against a market model) cross-check them.

5. Risk

Value-at-Risk and expected shortfall are computed conditional on the supply-shock state, using filtered historical simulation, and backtested with Kupiec and Christoffersen tests.

Thesis hook: let the satellite activity index a_t drive the jump intensity of a jump-diffusion, $\lambda_t = \exp(\theta_0 + \theta_1 a_t)$, and estimate it by MCMC.

Data, models and compute

Everything below is free. The binding constraint is compute, so cloud platforms do the heavy extraction and your laptop does modeling.

Satellite and GIS access

Source	What you get	Access
Copernicus Data Space Ecosystem	Sentinel-1, -2, -3, -5P via openEO, Sentinel Hub, STAC and S3	Free account; 10,000 openEO credits on the 1st of each month
Microsoft Planetary Computer	Public STAC API over Landsat, Sentinel-2 L2A and more, as cloud-optimized GeoTIFFs	Open search via `pystac-client`; read only AOI windows
NASA Earthdata	HLS 30 m surface reflectance; VIIRS night lights	Free Earthdata login

Source	What you get	Access
Google Earth Engine	AlphaEarth Satellite Embedding dataset (annual 64-dim embeddings); global mining polygons in the community catalog	Free for noncommercial and research use
EOG, Payne Institute	VIIRS Nightfire nightly flare detections (CSV, KMZ); annual global flare catalog	Free account required
USDA CroplandCROS	Cropland Data Layer; 2025 layer released Feb 27, 2026	Public domain
PANGAEA	Maus et al. global mining polygons: 21,060 polygons, 57,277 km²	Open download

Hugging Face

- Models: `ibm-nasa-geospatial/Prithvi-EO-2.0-300M-TL`, with `-100M-TL` and `-tiny-TL` for the laptop; `ibm-esa-geospatial/TerraMind-1.0-tiny`, `-small`, `-base` (multimodal, with S1, S2, land-cover and NDVI tokenizers); `made-with-clay/Clay`; `ibm-nasa-geospatial/Prithvi-EO-1.0-100M-multi-temporal-crop-classification` as a worked crop example.
- Datasets: `Major-TOM/Core-S2L2A`, `Core-S1RTC`, `Core-VIIRS-Nighttime-Light`, `Core-DEM` (global patches for backgrounds and pretraining); `ibm-nasa-geospatial/multi-temporal-crop-classification`.
- Publish back: your monthly activity indices as a dataset, plus a small Gradio Space with a map and index chart. That makes the work citable and visible beyond GitHub.

Your Claude connectors

- **Alpha Vantage:** monthly global copper, aluminum, corn and wheat prices; daily WTI, Brent, natural gas; yields, fed funds, CPI, payrolls, unemployment; `NEWS_SENTIMENT`; miner and ag equities. The monthly metal series lag: copper's latest print today is July 2026.
- **USAFacts:** FRED, BLS, EIA, Treasury, and USITC DataWeb for HTS-level imports and tariff rates, which become the edges of your supply network.
- **alphaXiv:** a literature folder plus question-answering over paper PDFs.

- **Hugging Face:** model and dataset search, and publishing your outputs.
- **Google Drive:** reference notes and exported figures.

Compute on the M3 Air (24 GB)

- Foundation models run inference only, on small AOI chips, with PyTorch MPS. Start with the tiny or 100M variants.
- Heavy temporal extraction runs on openEO or Earth Engine; never download full scenes.
- Keep AOIs small: three states, one basin, ~30 smelters, ~20 mines.
- Daily prices come from futures via yfinance (`ZC=F`, `CL=F`, `HG=F`); Alpha Vantage monthly series carry the long history.

Repo layout

Same conventions as the rest of the portfolio: numbered scripts behind one `bash run.sh`, Marimo notebooks, outputs committed, `.env` and `*.parquet` ignored. Each module runs the full pipeline with `--module corn|permian|copper`.

```
GIS-ML-Commodities-Network/
├── config/
│   ├── aois/                # corn_belt.geojson, permian.geojson,
│   │                        # smelters.geojson, mines.geojson,
ports.geojson
│   └── settings.yaml        # date ranges, release calendars, thresholds
├── pipelines/
│   ├── 01_ingest_aois.py
│   ├── 02_ingest_satellite.py      # STAC / openEO / Earth Engine pulls
│   ├── 03_ingest_market_macro.py   # Alpha Vantage, FRED, EIA, USITC, GPR
│   ├── 04_features_indices.py      # NDVI, flare heat, SWIR anomaly, NO2/SO2
│   ├── 05_embeddings.py            # Prithvi / TerraMind / AlphaEarth
│   ├── 06_nowcast_supply.py        # nowcast + supply surprise vs consensus
│   ├── 07_network.py               # supply graph, HHI, chokepoints
│   ├── 08_price_linkage.py         # regimes + state-dependent LPs
│   ├── 09_risk_scenarios.py        # conditional VaR/ES, scenarios
│   └── 10_report.py
├── r/                         # lpirfs + eventstudies scripts
├── src/atlas/                 # ingest/, features/, models/, network/, viz/
```

```
├── notebooks/                  # Marimo walkthroughs, one per module
├── app/                        # dashboard tab with maps
├── outputs/  figures/  reports/   # committed
└── run.sh
```

Build order

About 17 weeks at evening pace, one module finished before the next opens. Phase 2 alone is a complete, pushable portfolio piece if time runs short.

1. **Week 1, plumbing:** AOI GeoJSONs; accounts for Copernicus, Earthdata, Earth Engine and EOG; one map of every node. Ships scripts `01` – `03` and the atlas map in the README.
2. **Weeks 2–5, corn end to end:** HLS NDVI over CDL corn pixels → county yield nowcast (hierarchical Bayes and LightGBM) → surprise vs WASDE → futures response. Ships a working repo.
3. **Weeks 6–8, Permian:** Nightfire flare heat → EIA production nowcast → WTI and Waha basis. Ships an upstream supply feature for `energy-fundamentals-nowcast`.
4. **Weeks 9–12, copper:** smelter heat and SO2, mine change detection, the three 2025 event studies. Ships a copper activity index and a short note on what was and was not visible at Grasberg.
5. **Weeks 13–15, regimes, network, risk:** HMM regimes, state-dependent local projections across modules, supply graph from USITC flows, conditional VaR. Ships the cross-commodity results table.
6. **Weeks 16–17, publish:** Hugging Face dataset and Space, dashboard tab, and a 6–8 page research note that doubles as a thesis proposal draft.

Guardrails

The same honesty rules you already apply to the tightness engine and the lithium index, extended to satellite data.

- **Proxies, not tonnage.** An activity index measures observed heat, greenness or change. Label it as a proxy for production, never as production.
- **Point-in-time data.** USDA and EIA revise. Store each vintage, align features to release dates, and never train on numbers that were not public yet.

- **Small samples.** Monthly copper since 2019 is fewer than 100 observations. Label Bayesian outputs as decision-support estimates, not calibrated frequencies.
- **Missing looks.** Clouds and revisit gaps are not random. Model missingness explicitly and fall back to Sentinel-1 radar, which sees through cloud.
- **Blind spots.** Underground mines and enclosed smelters are partly invisible from orbit. Say so per node.
- **Many tests.** Dozens of nodes and signals invite false hits. Hold out the last 12 months, and apply the deflated Sharpe ratio to any trading rule.

Sources

Data and platforms

- Copernicus Data Space Ecosystem: openEO credits
- Microsoft Planetary Computer data catalog
- AlphaEarth Foundations and the Satellite Embedding dataset (Google DeepMind, July 2025)
- VIIRS Nightfire and the global gas flare catalog (EOG, Payne Institute)
- USDA NASS Cropland Data Layer; CDL as a cloud-optimized Zarr store, 30 m and 10 m
- Maus et al. 2020, A global-scale data set of mining areas; data on PANGAEA; Earth Engine community catalog copy
- Caldara–Iacoviello Geopolitical Risk Index
- Earth-i SAVANT smelter index FAQ

Models and datasets on Hugging Face

- Prithvi-EO-2.0-300M-TL
- TerraMind-1.0-base
- Clay
- Major-TOM Core-S2L2A
- Multi-temporal crop classification dataset

2025 copper disruptions

- Freeport declares force majeure at Grasberg (Project Blue)
- Copper price reaction to the force majeure (Oregon Group)
- Kakula seismic flooding and Grasberg (Livewire)

Papers to start with

- Watching Trade from Space: measuring maritime trade with satellite imagery (2026; SAR plus night lights, public data only)
- MineNetCD: global mining change detection benchmark (2024)
- AgroBench: crop yield learning from county statistics and pixels (2026)